

Introduction to Cryptocurrencies, Fundamental and Technical Analysis

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Introduction to Crypto Currencies & Tokenomics

This section aims to introduce the broader cryptocurrency landscape beyond just Bitcoin. The goal is to build an understanding of how different tokens are structured, what problems they attempt to solve, and what gives them value. Here we introduce to the concept of Tokenomics: the study of a token's supply, distribution, incentives, and utility, as a way of assessing sustainability and design.

Key concepts covered in this section:

- Use case, demand, and scarcity — why the token exists and how it's positioned in the market
- Total supply vs circulating supply — how many tokens exist overall, and how many are tradable or released
- Inflationary vs deflationary models — whether the token's supply increases over time or decreases through burns or caps
- Staking mechanisms — users lock up tokens to earn rewards/secure the network
- Burning — permanently removing tokens from circulation to reduce supply
- Vesting schedules and unlock periods — how and when tokens are released to team members, investors, or communities

Other Relevant Cryptocurrencies and Tokens

In addition to privacy coins, oracle tokens, functional coins (like XRP), and Layer 1's (such as ADA), there are several other categories of cryptocurrencies worth knowing. Here is a short breakdown of the different types:

- Bitcoin (BTC) and Layer 1s - Bitcoin (BTC) -The original cryptocurrency, designed as a decentralized digital currency and store of value.
- Layer 1s (Ethereum, Solana, Avalanche) - Base blockchains that provide the underlying infrastructure for transactions, applications, and smart contracts.
- Layer 2s - Built on top of Layer 1s to increase scalability, speed, and reduce costs (e.g., Polygon, Arbitrum, Optimism for Ethereum). They offload transaction processing from the main chain while maintaining its security.
- Layer 3s -Application-specific protocols built on top of Layer 2s, enabling specialized use cases like gaming, DeFi, or data storage with more customization.
- Oracle Tokens -Tokens like Chainlink (LINK) that provide real-world data feeds (prices, weather, events) to smart contracts, bridging blockchains with external information.

- Privacy Coins -Cryptocurrencies designed to maximize transaction anonymity (e.g., Monero, Zcash), obscuring sender, receiver, or amounts from public view
- Stablecoins – Digital assets pegged to stable values (usually USD, EUR, or commodities). They reduce volatility and are often used for trading pairs and payments. Examples: USDT (Tether), USDC, DAI.
- Governance Tokens – Tokens that give holders voting rights over protocol changes, treasury allocation, or ecosystem decisions. They are central to decentralised governance. Examples: UNI (Uniswap), COMP (Compound).
- Meme Coins – Highly speculative tokens driven by community hype, internet culture, or humor. While risky, some achieve large market caps due to viral attention. Examples: DOGE (Dogecoin), SHIB (Shiba Inu).
- CBDCs (Central Bank Digital Currencies) – Digital currencies issued directly by governments or central banks. Unlike cryptocurrencies, they are centralized but use blockchain-inspired technology. Examples: e-CNY (China’s digital yuan), proposed digital Euro, or the Bahamas’ Sand Dollar.

Token Deep Dive & The Blockchain Trilemma

Oracle Networks Blockchains on their own cannot access real-world data (like prices, weather, or event outcomes). Oracle networks solve this by acting as bridges between blockchains and external information.

Technical Features:

- Decentralised Data Feeds – Instead of one source, oracle networks aggregate inputs from multiple providers to prevent manipulation or downtime.
- Price Feeds – Secure, real-time updates for assets (e.g., ETH/USD) used by DeFi protocols for lending, borrowing, and trading.
- Verifiable Randomness (VRF) – Provides provably fair random numbers for applications like gaming, NFTs, and lotteries.
- Cross-Chain Communication – Some oracle systems enable messages and assets to move between different blockchains, improving interoperability.
- Automation (Keepers) – Smart contracts can be set to trigger actions automatically when conditions are met (e.g., liquidating a position if collateral drops).

Chainlink (LINK) Is the most widely use oracle network that brings real-world data (like prices, weather, events) onto blockchains. It’s essential for smart contracts that need external data — especially in DeFi. LINK tokens pay node operators for accurate and reliable data.

Monero (XMR) vs Zcash (ZEC)

Both are privacy coins, but they achieve privacy in different ways:

- Monero (XMR) – Privacy is enabled by default. It hides the sender, receiver, and amount using features like ring signatures, stealth addresses, and RingCT (Ring Confidential Transactions).
- Zcash (ZEC) – Privacy is optional. Users can choose between transparent transactions (like Bitcoin) and shielded transactions, which use advanced zk-SNARKs (zero-knowledge proofs) to conceal details.

Technical Features / Key Differences:

- Default vs Optional Privacy: Monero enforces privacy for all transactions, while Zcash leaves it up to the user.
- Blockchain Transparency: Zcash still has visible transparent addresses, while Monero is fully obfuscated.
- Cryptography: Monero relies on ring signatures and stealth addresses, while Zcash uses zk-SNARKs, which are computationally heavier but allow flexible privacy.

XMR vs BTC (Cash vs Gold):

Design Philosophy:

- Bitcoin (BTC) is often seen as digital gold — a store of value with transparent, auditable transactions.
- Monero (XMR) functions more like digital cash — private, fast, and fungible for everyday use.

Proof-of-Work Differences:

- BTC uses SHA-256, which favours ASIC miners and centralises hashing power into large mining farms.
- XMR uses RandomX, an algorithm designed to be ASIC-resistant and CPU-friendly, keeping mining more decentralised.

Speed & Block Times:

- BTC: ~10-minute block time → slower confirmations.
- XMR: ~2-minute block time → faster, more practical for small payments.

Fungibility:

- BTC coins can be “tainted” by history (traceable through block explorers).
- XMR coins are indistinguishable from one another, making them fully fungible like physical cash.

Summary:

- Monero = privacy + speed + everyday use (digital cash).
- Bitcoin = transparency + security + long-term store of value (digital gold).
- Zcash = a hybrid approach, balancing optional privacy with transparency.

Layer 1's Layer 1 blockchains are the foundational networks on which applications and other protocols are built. Each Layer 1 makes trade-offs between speed, decentralisation, and security — the core of the blockchain trilemma.

Solana vs Ethereum

- Solana (SOL): A high-speed, low-fee Layer 1 designed for scalability. It uses a unique consensus innovation called Proof of History (PoH), which timestamps transactions before they are processed. This acts like a cryptographic clock, allowing validators to agree on the order of events without waiting on one another. The result is high throughput (50k+ TPS) and low latency. However, Solana has faced criticism for centralisation risks (fewer validators, hardware requirements) and occasional network outages.
- Ethereum (ETH): The most widely adopted Layer 1, hosting the largest smart contract and DeFi ecosystem. Recently transitioned from Proof of Work to Proof of Stake (the Merge), improving energy efficiency and scalability potential. Ethereum is slower and more expensive in terms of fees, but it is more decentralised and secure, with the deepest developer and user community.

Other Major Layer 1's

- Cardano (ADA): Built with a strong academic and peer-reviewed approach, using its own Proof of Stake protocol called Ouroboros. Known for its focus on sustainability, modular upgrades, and scalability.
- Avalanche (AVAX): Uses a novel consensus mechanism designed for near-instant finality and very high throughput, with a flexible subnet structure for customised blockchains.
- Polkadot (DOT): Aims to enable cross-chain interoperability with a hub-and-spoke model of parachains connected to a central Relay Chain.

Summary:

- Ethereum = secure, decentralised, largest ecosystem.
- Solana = ultra-fast, low-cost, but more centralised.
- Cardano = sustainable, research-driven, modular Proof of Stake.

- Avalanche/Polkadot = pushing scalability and interoperability.

XRP

Although not a Layer 1 in the same sense, XRP is worth noting as a functional token designed for fast, low-cost cross-border payments. It operates on the XRP Ledger (XRPL), which processes transactions in $\tilde{3}$ –5 seconds with minimal fees, targeting financial institutions and remittances.

Used by Ripple for fast, low-cost cross-border payments. XRP doesn't use mining — instead, it relies on a trusted validator set. It's scalable and efficient but faces criticism over centralisation due to Ripple's control of the network and token supply.

Consensus Mechanisms (Revisited)

- Proof of Work (PoW) — Bitcoin: secure, decentralised, energy-intensive.
- Proof of Stake (PoS) — Ethereum: lower energy, faster block times.
- Proof of History (PoH) — Solana: optimised for speed and throughput.
- Unique Node List (UNL) — XRP: fast consensus via trusted validators.

The Blockchain Trilemma

Different blockchains prioritise different things:

- Bitcoin → decentralisation + security (slow but stable).
- Solana → scalability + speed (more centralised).
- Ethereum → aiming for balance.
- XRP → scalability, but less decentralised.
- Monero/Zcash → privacy and decentralisation over performance.

Fundamental Analysis (FA)

This section focuses on how to assess a cryptocurrency project's long-term value and overall sustainability. FA involves looking at the actual foundations of a project: who built it, why it exists and how it works, rather than just its price movements.

Cryptocurrency project meaning:

- The team or organisation building it (e.g Solana Labs)
- The problem its trying to solve (fast payments, decentralised, lending, storage, privacy)
- The product or protocol (blockchain, app, smart contracts)
- The community and developers using or contributing to it

- What is the token itself used for? Governance transactions, staking etc.
- The roadmap and updates i.e. where it's headed in the future.

In time, we will explore how to examine the founding team and advisors behind a project. This includes researching whether the creators are credible, whether they are anonymous or public, and whether they have previous experience in tech, blockchain, or business.

The use case of the project is also key as we should ask ourselves what real-world or blockchain problem it is solving, and whether there is demand for that solution. Tokenomics plays a central role here.

- We will review things like the token's max supply, how it's distributed, and whether it's inflationary or deflationary.
- We also consider staking, liquidity incentives, and emissions — how tokens are rewarded over time.

The roadmap and development activity should also be assessed.

- This includes looking at GitHub activity, regular updates, and how active the dev team is in delivering progress.

Community strength is another important factor.

- A strong community may include active Discord channels, engaged Twitter followers, DAO activity, and educational content created by users.

You should also explore whether the project is gaining adoption.

- This can include integrations with other protocols, partnerships with major platforms or mentions in industry media.

Bitcoin – Debates, Design and Criticisms

Key Technical Features of Bitcoin

Bitcoin is a decentralised, peer-to-peer digital currency designed as an alternative to centralised fiat systems. Its core innovations include:

- Cryptographic hash functions, digital signatures, and timestamped append-only logs (blocks), secured using asymmetric encryption.
- The consensus mechanism Proof of Work (PoW), where miners compete to solve puzzles, adding new blocks to the chain.
- It uses a UTXO (Unspent Transaction Output) model, where transactions are composed of inputs and outputs, tracked as discrete units rather than balances.
- Bitcoin is not Turing complete: it uses a limited scripting language for simplicity and security.

- Its network is secured by miners using ASICs (Application-Specific Integrated Circuits) with a hash rate exceeding 50 Exahash/s. Tech improvement has resulted in this, and made block mining 10 trillion times harder.

Block time averages 10 minutes, and its supply is hard-capped at 21 million coins, halving roughly every 4 years. Fees are voluntary but incentivise miners.

Bitcoin vs Ethereum: Key Differences

- Ethereum is Turing complete — supporting smart contracts written in Solidity and other languages.
- It uses an account-based model (not UTXO) with state transitions between balances.
- While Bitcoin remains on PoW, Ethereum has moved to Proof of Stake.
- Ethereum blocks are faster (~12 seconds), and gas fees are based on demand via its EIP-1559 upgrade.

BTC is optimised for simplicity, security, and monetary use cases. ETH is optimised for flexibility and programmability.

Bitcoin Criticisms and Maximalist Counterpoints

1. Volatile, not a reliable store of value
Critique: BTC is too volatile for saving or pricing goods.
Counter: Scarcity and halving cycles drive long-term value. Like gold, volatility is expected during early adoption phases.
2. Over-reliance on centralised exchanges
Critique: Exchanges hold user funds and can be hacked.
Counter: Self-custody via wallets removes this risk. Bitcoin itself is decentralised — exchanges are just access points.
3. 51% attack risks
Critique: A miner with majority hash power can rewrite history.
Counter: Bitcoin is the most secure network ever created — attacking it is prohibitively expensive.
4. Mining consumes too much energy
Critique: BTC mining is unsustainable.
Counter: Security and decentralisation require energy. Innovations like hydro/solar mining and Layer 2 (Lightning) improve efficiency.
5. Lacks scalability and performance
Critique: Transactions are slow and expensive.
Counter: Bitcoin is a base settlement layer. Lightning Network and protocol upgrades like SegWit and Taproot enable faster, cheaper payments.

6. No intrinsic value
Critique: BTC is not backed by anything.
Counter: Its value lies in digital scarcity, censorship resistance, and borderless monetary utility.
7. Low adoption for real-world payments
Critique: Most people don't use BTC to buy things.
Counter: Adoption is growing (El Salvador, global remittances, institutional holdings). Progress is slow, but steady.
8. Hardcoded monetary policy
Critique: 21M cap and halving aren't flexible.
Counter: Predictable supply is a strength. It removes inflation risk and political manipulation.
9. Prone to centralisation (mining pools)
Critique: Mining power is concentrated.
Counter: Mining pools are dynamic and globally distributed. Bitcoin's incentives discourage long-term centralisation.
10. Lose keys, lose funds
Critique: No account recovery.
Counter: Financial sovereignty includes responsibility. Multi-signature wallets and custodians offer safer options.
11. No 'killer app'
Critique: BTC has no dominant use case beyond speculation.
Counter: Bitcoin *is* the app meaning it is secure, decentralised, censorship-resistant money.
12. Crypto scams give BTC a bad name
Critique: The space is full of fraud.
Counter: Most scams occur in altcoins. Bitcoin's protocol has never been hacked and is transparent by design.